Project: Monte Carlo simulation of solid crystalline phases

Objective: To study the structure of a solid system in different crystalline phases.

Outline

1. Write a MC code for a (N,V,T) simulation of a Lennard-Jones (LJ) system. Use N $\leq$ 500 atoms. Remember that the LJ potential is of the form: $U_{LJ}(r) = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^{6}]$.

2. Create two initial set-ups of positions:

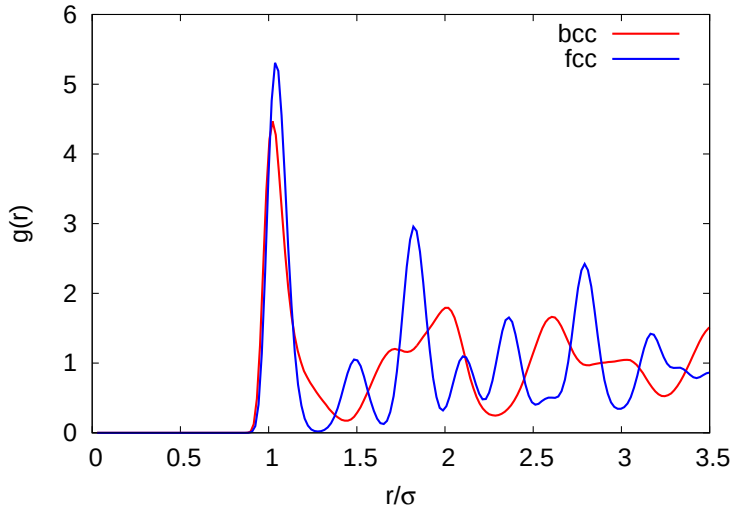
   I) from the sites of a bcc (*body centred cubic*) lattice. (HINT: consider a bcc lattice as the superposition of 2 single cubic lattices.)

   II) from the sites of a fcc (*face centred cubic*) lattice. (HINT: consider a fcc lattice as the superposition of 4 single cubic lattices.)

3. For each of the initial set-ups, run a MC simulation for solid argon at T* = 1.0 and ρ^*= 1.2. Compute the interaction energy U^*, the pressure P^*, and the radial distribution functions $g(r)$.

4. Repeat the simulations for the following temperatures: T*= 2.0, T*=3.0 and T*=4.0, and compare with the previous results.

Some results to compare with:



Radial distribution functions for the different crystalline phases at T*=1.0 .

Questions

1. **Summarize the Metropolis algorithm for (N,V,T) simulations. Describe the main hypothesis involved in its derivation.**

2. Define the reduced units used in your simulations. Translate from reduced units to real units the values of T^*, ρ^*, U^* and P^*.

3. Give an interpretation of the peak positions shown by the $g(r)$ functions at T*=1.0. Compute for each system the so-called coordination number n_c, i. e. the mean number of atoms in the first coordination shell.

4. Discuss for both crystalline phases the differences you find in the $g(r)$ when heating the system.